Lab Problem 6

Objective:

To implement a **Generalized Linear Model (GLM)** using a **Bernoulli distribution** and **Logistic Sigmoid link** to demonstrate how the Exponential Family framework provides robustness against spatial outliers.

Dataset Description

- You will work with a synthetic 2D dataset (N=51):
- **Class 0:** 25 samples centered at (2, 2).
- **Class 1:** 25 samples centered at (-2, -2).
- **The Outlier:** A single Class 0 observation located at (10, 10).

Tasks:

You must perform the following tasks in order:

Task 1: Feature Augmentation

- **To incorporate the bias term**(w_0) **transform the input features** $\mathbf{x} = [x_1, x_2]^T$ **into the augmented vector:**

$$\phi = [1, x_1, x_2]^T$$

Task 2: The Inverse Link Function

- Implement the **Logistic Sigmoid** function to map the linear predictor a to the predicted probability y :

$$y = \sigma(a) = \frac{1}{1+e^{-a}}$$

Task 3: Gradient Descent Optimization

- Using the Maximum Likelihood principle for the Exponential Family, find the optimal weight vector $\mathbf{w}$ by implementing the following update rule:

$$\mathbf{w_{new}} = \mathbf{w_{old}} - \eta \sum_{n=1}^{N} (y_n - t_n)\phi_n$$

- **Parameters:** Learning rate η = 0.1, Iterations = 1000.

Visualization & Analysis

- **Decision Boundary:** Plot the data and the boundary line where $w_0 + w_1 x_1 + w_2 x_2 = 0$
- **Analysis:** Explain how the "squashing" property of the sigmoid function prevents the outlier at (10, 10) from shifting the boundary, unlike a Least Squares model.